Huawei Access Node BAS-ACC-07 Optical Transport Failure Mitigation SOP for Basavanagudi POP

Node: BAS-ACC-07 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_5774

Description

This Standard Operating Procedure (SOP) outlines the steps to mitigate a Major Optical Transport Failure on Huawei Access Node BAS-ACC-07 at Basavanagudi POP, triggered by an Alarm Code ALARM_5774. The failure domain, characterized by SFP/XFP degradation, fiber cuts, or Rx/Tx power loss, affects the Optical Transport module responsible for transmitting data between nodes in the network. This module relies on the Optical Transport Protocol (OTP) to manage the data transmission and reception via SFP/XFP transceivers. However, hardware or environmental malfunctions can lead to a degradation in the transceivers' performance, resulting in the failure of the Optical Transport module. Furthermore, the failure of the Optical Transport module can lead to a cascade of failures in the Access Node, disrupting the transmission of data packets and significantly impacting network performance. To mitigate this failure, a structured approach is required, involving the identification of the root causes, execution of diagnostic commands to isolate the issue, and implementation of resolution steps to rectify the problem. This SOP aims to provide a step-by-step guide for the network engineer to follow in order to effectively mitigate the failure and restore normal network operations.

Pre-requisite Checks

- Verify that the network engineer has access to the Huawei Access Node BAS-ACC-07 at Basavanagudi POP.

- Ensure that the redundant link is available and functioning properly to prevent network disruption during the repair process.

- Check the environmental conditions at the POP, including temperature, humidity, and power supply, to ensure they are within the normal operating range.

- Notify the NOC (Network Operations Center) of the failure and obtain their approval to proceed with the repair.

Potential Root Causes

Cause	Probability
SFP/XFP degradation due to aging or environmental factors.	High
Fiber cuts or damage due to physical factors.	Medium
Rx/Tx power loss due to misconfigured or faulty optical fibers.	Low

Diagnostic Commands & Expected Outputs

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Command: display optic-fiber interface, Expected Output: Optic-fiber interface information, Description: This command displays the status of the optic-fiber interface, including the received and transmitted power levels, which can help identify if there is a power loss issue.

Command: display optic-fiber link, Expected Output: Optic-fiber link information, Description: This command displays the status of the optic-fiber link, including the signal-to-noise ratio (SNR) and the error rate, which can help identify if there is a fiber cut or degradation issue.

Command: display optic-fiber alarm, Expected Output: Optic-fiber alarm information, Description: This command displays the alarm information related to the optic-fiber, including the alarm code and description, which can help identify the root cause of the failure.
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Resolution Steps

Step 1: Step: Replace the faulty SFP/XFP transceiver, Command: replace optic-fiber interface , Description: Replace the faulty SFP/XFP transceiver with a new one to rectify the issue. Ensure that the new transceiver is properly installed and configured.

Step 2: Step: Repair or replace the damaged fiber, Command: repair optic-fiber link , Description: Repair or replace the damaged fiber to rectify the issue. Ensure that the repaired or replaced fiber is properly installed and configured.

Step 3: Step: Reconfigure the Rx/Tx power settings, Command: reconfigure optic-fiber interface rx-tx-power, Description: Reconfigure the Rx/Tx power settings to rectify the issue. Ensure that the new power settings are properly configured and verified.

Post-Check Verification

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Command: display optic-fiber interface, Expected Output: Optic-fiber interface information, Description: Verify that the optic-fiber interface is functioning properly and that the power levels are within the normal operating range.

Command: display optic-fiber link, Expected Output: Optic-fiber link information, Description: Verify that the optic-fiber link is functioning properly and that the SNR and error rate are within the normal operating range.

Command: display optic-fiber alarm, Expected Output: Optic-fiber alarm information, Description: Verify that the optic-fiber alarm is cleared and that there are no further alarm triggers.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP transceiver, Command: restore optic-fiber interface , Description: Restore the original SFP/XFP transceiver if the replacement does not resolve the issue.

- Step: Restore the original fiber, Command: restore optic-fiber link , Description: Restore the original fiber if the repair or replacement does not resolve the issue.

- Step: Restore the original Rx/Tx power settings, Command: restore optic-fiber interface rx-tx-power, Description: Restore the original Rx/Tx power settings if the reconfiguration does not resolve the issue.